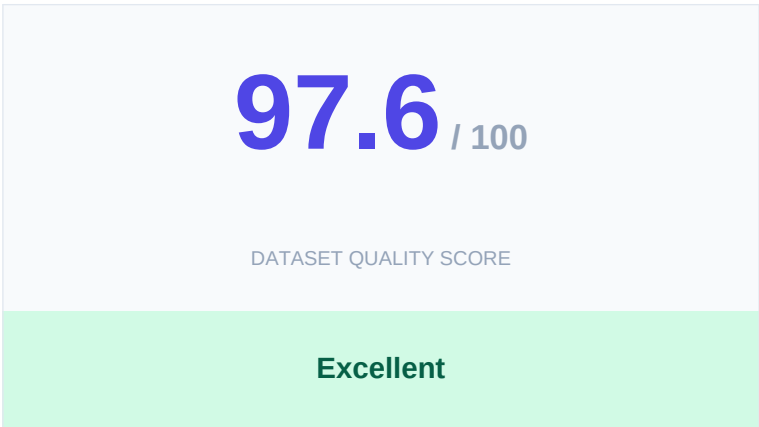


EXPLAINABLE AUTOEDA

Comprehensive Data Intelligence Report

Generated on 04 May 2026 at 01:17

[Data Cleaning](#) · [EDA](#) · [Quality Scoring](#) · [Preprocessing](#)



Metric	Original	Cleaned
Rows	312	300
Columns	15	15
Missing Cells	136	0
Duplicate Rows	12	0

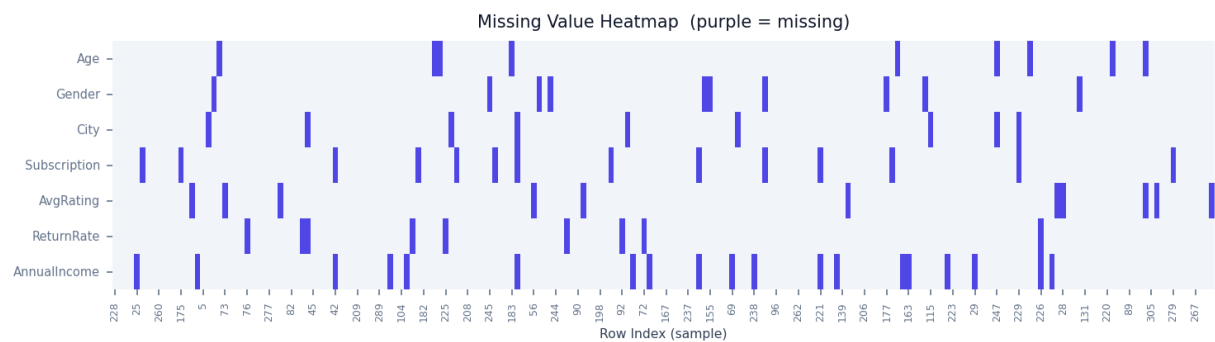
01 Quality Score Breakdown

- ✓ Missing values: 0-pt penalty — no missing cells detected.
- ✓ Duplicate rows: 0-pt penalty — no duplicates detected.
- Outliers: -1.4 pts (1 of 14 numeric column(s) contain IQR outliers). Fix: apply capping or removal in Step 03 → Outlier Detection & Treatment.
- Unencoded categoricals: -1.0 pts (1 text/category column(s) still unencoded: Name). Fix: apply Label or One-Hot encoding in Step 03 → Categorical Encoding.
- ✓ Type consistency: 0-pt penalty — all columns have consistent types.
- ✓ Column completeness: 0-pt penalty — all columns contain at least one value.

SCORE METHODOLOGY

- Formula: 100 – penalties across 6 dimensions.
- Missing values: up to 30-point penalty based on average missing rate.
- Duplicate rows: up to 20-point penalty based on duplicate row percentage.
- IQR Outliers: up to 20-point penalty for numeric columns with outliers.
- Unencoded categoricals: up to 15-point penalty for text/category columns.
- Type inconsistency: up to 10-point penalty for mixed-type columns.
- Empty columns: up to 5-point penalty for entirely empty columns.
- Rating scale: ≥90 Excellent | ≥75 Good | ≥50 Fair | <50 Poor.

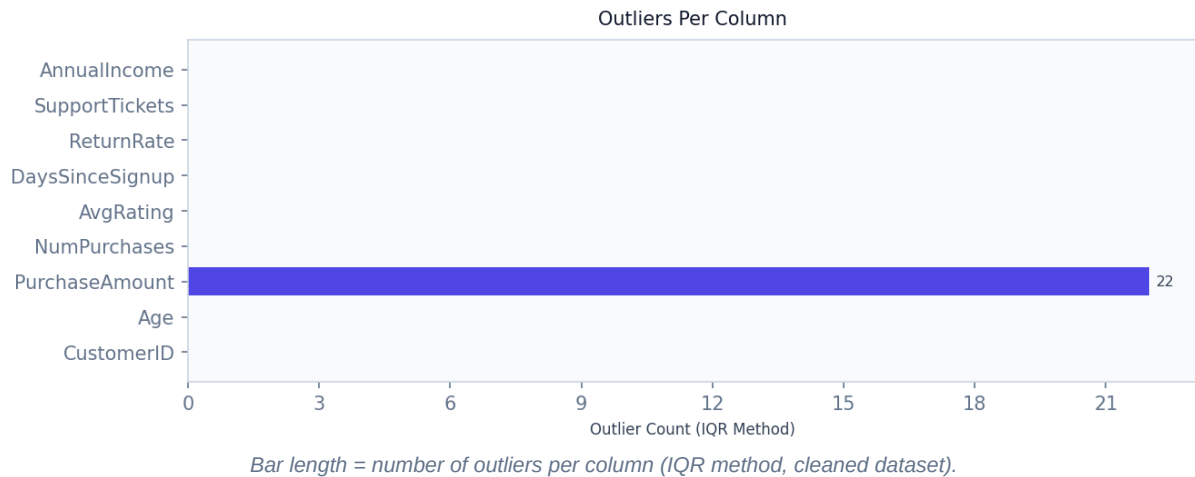
02 Missing Value Analysis



MISSING VALUE SUMMARY

- Total missing cells: 136 (2.91% of all data).
- Affected columns (7): Age, Gender, City, Subscription, AvgRating, ReturnRate, AnnualIncome.
- Most affected column: AnnualIncome (9.0% missing).
- Numeric columns: Median imputation (robust to outliers).
- Categorical columns: Mode imputation (preserves most frequent value).
- Columns with >30% missing values warrant careful review — consider dropping them.

03 Outlier Detection



OUTLIER SUMMARY

- Method: IQR — values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are flagged.
- Columns with outliers: PurchaseAmount: 22
- Capping (Winsorization): replaces outliers with boundary values — preserves row count.
- Row removal: only appropriate for confirmed data entry errors.
- Outliers inflate standard deviation and distort regression coefficients.

04 Column Statistics (cleaned dataset)

Column	Mean	Median	Std Dev	Min	Max	Outliers
CustomerID	1150.500	1150.500	86.747	1001.000	1300.000	0
Age	0.497	0.510	0.289	0.000	1.000	0
PurchaseAmount	0.320	0.228	0.293	0.000	1.000	22
NumPurchases	0.315	0.299	0.206	0.000	1.000	0
AvgRating	0.509	0.525	0.283	0.000	1.000	0
DaysSinceSignup	0.489	0.474	0.276	0.000	1.000	0
ReturnRate	0.496	0.480	0.291	0.000	1.000	0
SupportTickets	0.467	0.444	0.315	0.000	1.000	0
AnnualIncome	0.504	0.512	0.195	0.000	1.000	0

STATISTICAL NOTES

- Mean vs Median: large gap = skewed distribution; use median as central measure.
- Std Dev: high relative to mean = high variability across records.
- Min / Max: verify against expected domain bounds — extremes may be data errors.
- Skewed columns ($|skew| > 1$): PurchaseAmount (skew=1.13)
- Highly skewed columns may benefit from log or Box-Cox transformation before modelling.

05 Correlation Analysis



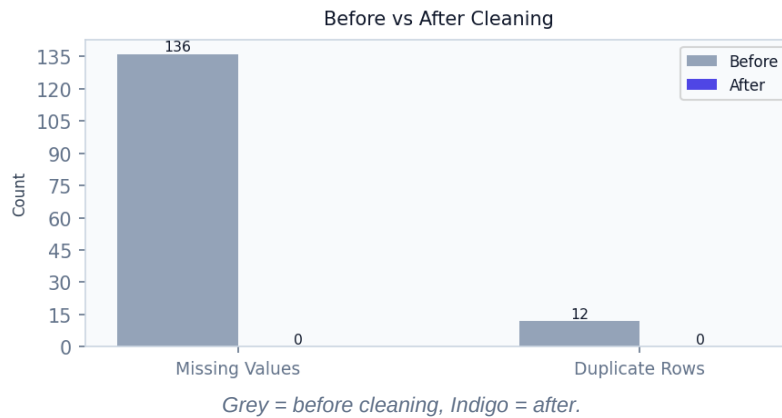
Green = positive, Red = negative. Values range from -1 (perfect inverse) to $+1$ (perfect direct). Lower triangle shown.

CORRELATION NOTES

- Pearson r measures linear association between numeric column pairs.
- Strength: $|r| \geq 0.80$ Strong | $0.50-0.79$ Moderate | $0.30-0.49$ Weak | <0.30 Negligible.
- Strong pairs ($|r| \geq 0.70$): None found.
- High correlation (multicollinearity) inflates coefficient variance in regression models.
- Pearson only captures linear relationships — supplement with scatter plots.

06 Before vs After Cleaning

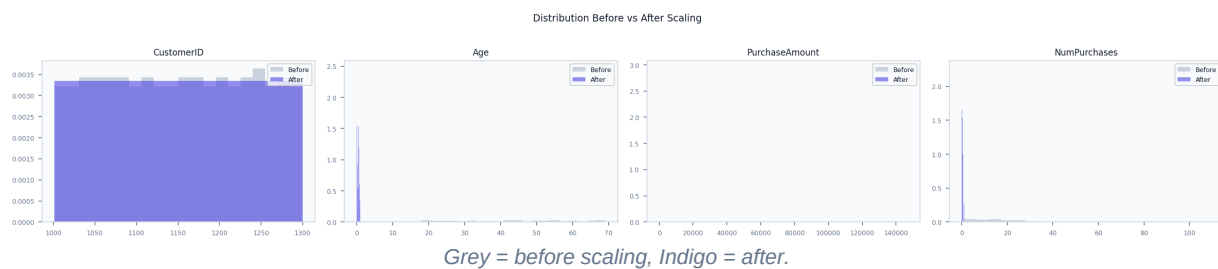
Metric	Before	After	Change
Rows	312	300	-12
Columns	15	15	—
Missing Values	136	0	-136
Duplicate Rows	12	0	-12
Missing %	2.91%	0.00%	



CHANGES APPLIED

- 136 missing values filled via imputation.
- 12 duplicate rows removed.
- Row count reduced from 312 to 300 (outlier or duplicate removal).
- All modifications reflect user-approved actions only.

07 Feature Scaling



SCALING NOTES

- Scaling brings numeric columns to a common range so no column dominates by magnitude alone.
- Min-Max Scaling: maps values to [0, 1]. Best for KNN, K-Means, and neural networks. Sensitive to outliers.
- Standard Scaling: transforms to mean=0, std=1. Robust to outliers. Required for PCA and linear models.
- Apply scaling after outlier treatment — outliers significantly distort Min-Max bounds.
- ID and name columns are automatically excluded from scaling.

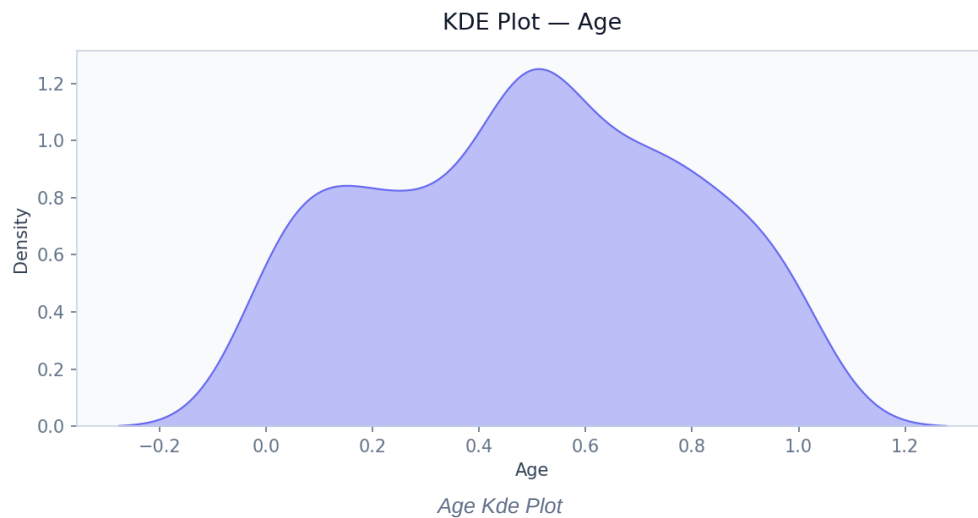
08 Categorical Encoding

Column	Action	Result
Category	Label Encoding	Integer codes
City	Label Encoding	Integer codes
Subscription	Label Encoding	Integer codes
Segment	Label Encoding	Integer codes
Gender	Label Encoding	Integer codes

ENCODING NOTES

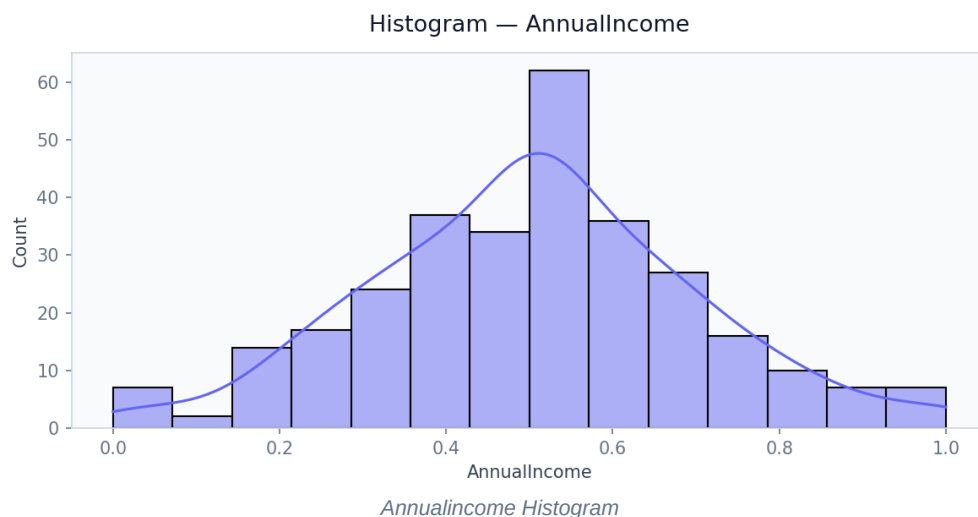
- Encoding converts text/category columns to numbers — required by most ML algorithms.
- Label Encoding: assigns an integer per category (e.g. Red=0, Blue=1). Best for ordinal data.
- One-Hot Encoding: creates one binary column per category. Best for nominal data (no natural order).
- Avoid One-Hot for columns with >10 unique values — it creates too many columns (curse of dimensionality).
- ID and name columns (e.g. CustomerID, Name) are automatically excluded from encoding.

09 Exploratory Data Analysis



KDE PLOT — NOTES

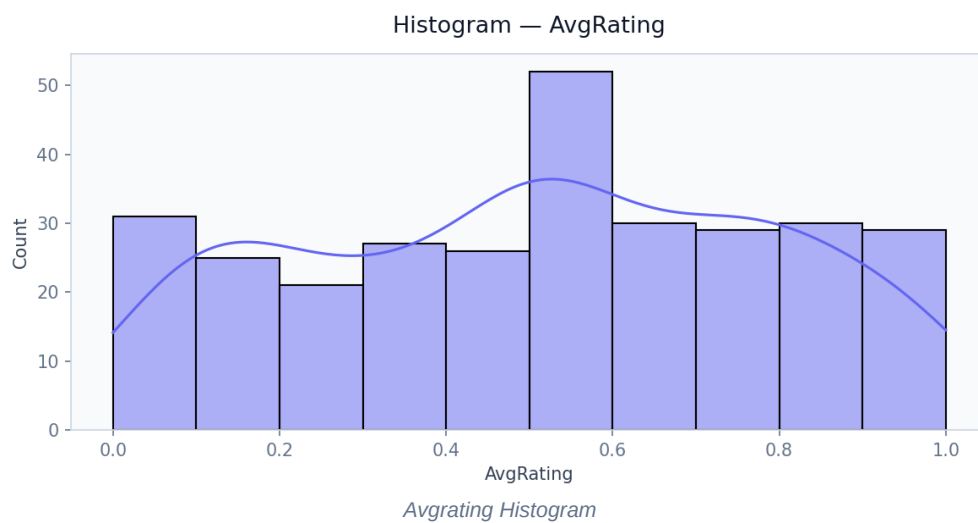
- Smoothed probability density — no bin-size artefacts.
- Multiple peaks suggest distinct sub-groups exist.



HISTOGRAM — NOTES

- Groups values into bins — reveals distribution shape.
- KDE curve overlaid to smooth bin-level noise.

- Long right tail = positive skew; watch for data entry errors.



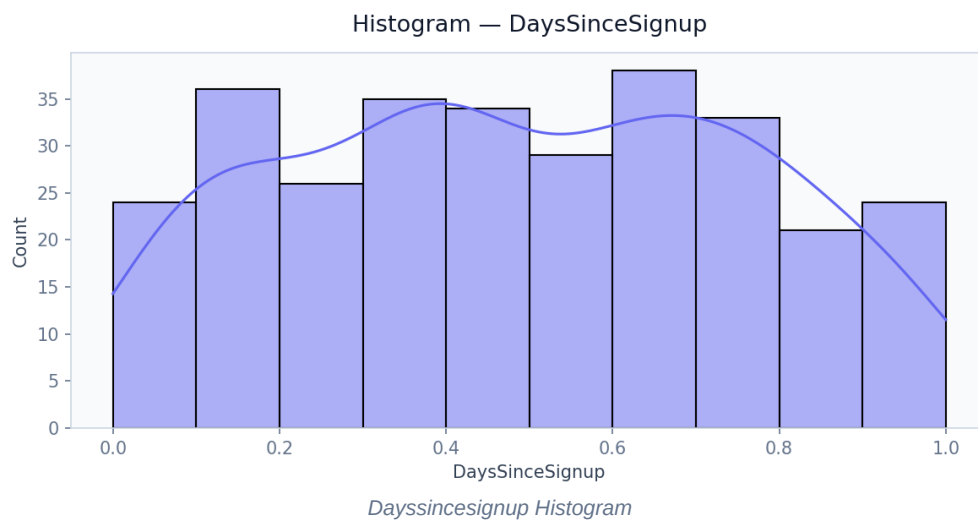
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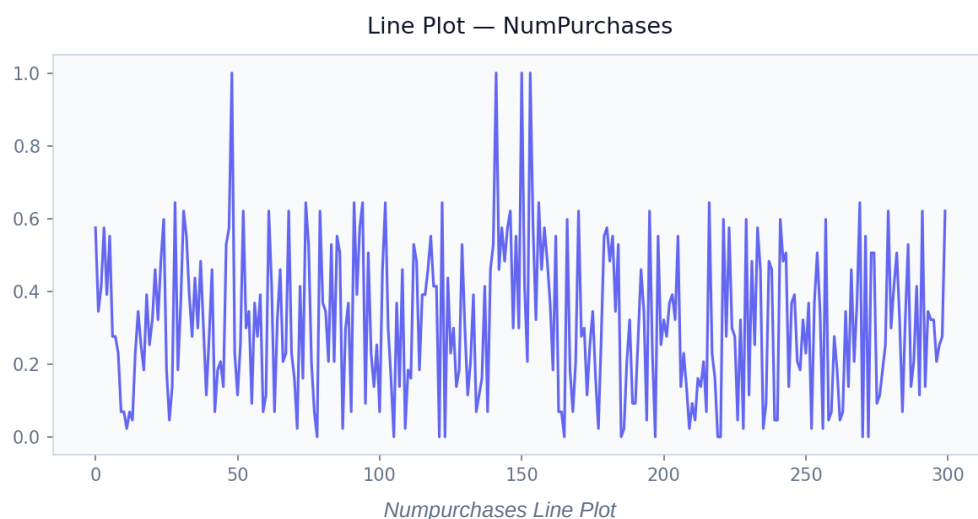
BOX PLOT — NOTES

- Shows Min, Q1, Median, Q3, Max in one view.
- Points beyond the whiskers are IQR-flagged outliers.
- Box shifted toward Q1 indicates right-skewed data.



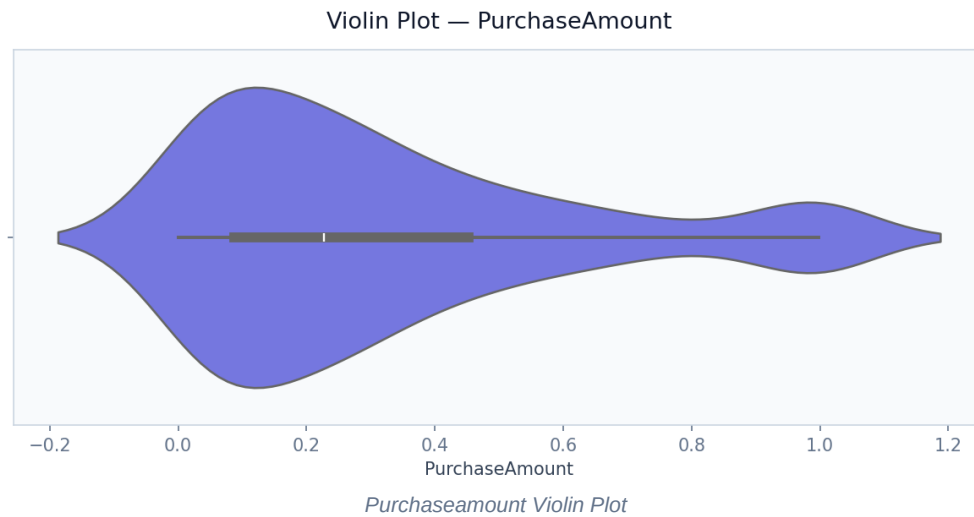
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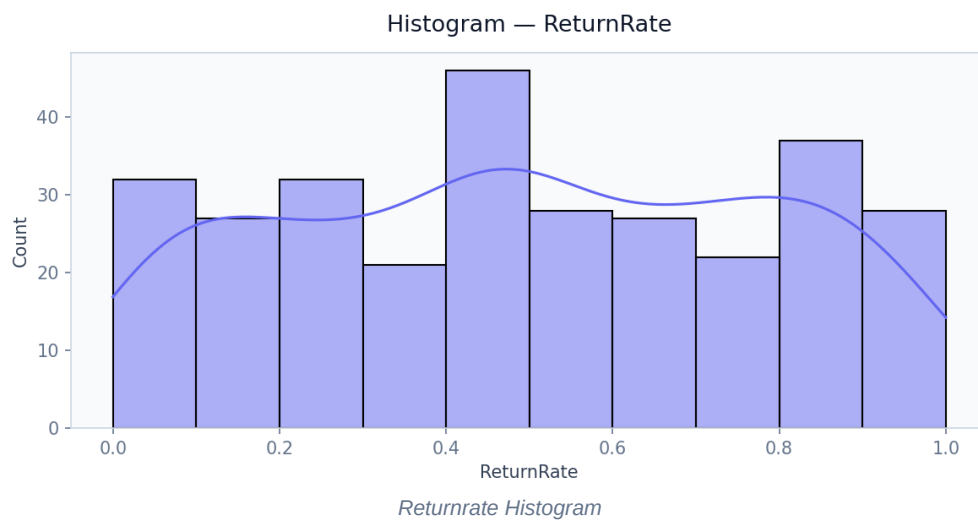
LINE PLOT — NOTES

- Plots values in row order — best for sequential/time data.
- Spikes or flat regions can signal anomalies or batch effects.



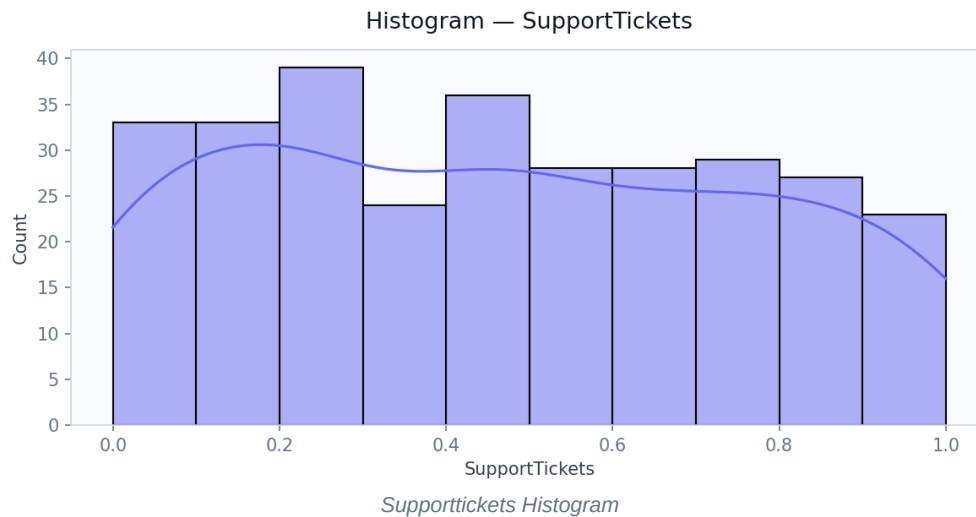
VIOLIN PLOT — NOTES

- Box plot + KDE combined — shows full density shape.
- Wide sections = dense value clusters.



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10 Cleaning Actions Summary

Cleaning actions were applied based on user approval.

ACTIONS LOG

- Imputed 136 missing values using median (numeric) / mode (categorical).
- Removed 12 duplicate rows.
- Median preferred over mean: not distorted by outliers.
- Duplicate removal prevents inflated counts and biased model training.
- ID and name columns were excluded from scaling and encoding.
- Suggested pipeline: feature engineering → train/test split → model selection → evaluation.

11 Recommendations & Next Steps

- Outliers remain in: PurchaseAmount. Consider capping or log-transform.
- 1 categorical column(s) unencoded: Name. Encode before modelling.
- Highly skewed: PurchaseAmount. Consider log or Box-Cox transformation.
- ✓ Quality score 97.6/100 (Excellent) — dataset is in good shape for analysis.
- ✓ Next steps: feature engineering → train/test split → model selection → evaluation.
- ✓ Save the cleaned CSV and document all preprocessing decisions for reproducibility.

This report was automatically generated by the Explainable AutoEDA Platform. All findings are based on the uploaded dataset and user-approved preprocessing actions.